

WHO CHOOSES AND WHO BENEFITS? THE DESIGN OF PUBLIC SCHOOL CHOICE SYSTEMS

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U.S. school districts have expanded public school choice through magnet, dual-language, and themed programs. Using original data on the 150 largest districts, we show that most assign students to these programs using coordinated, single offer systems but leave participation optional. These opt-in systems are harder to navigate than mandatory systems and disproportionately serve advantaged students. We study Los Angeles's opt-in system, the nation's largest, using administrative records, lotteries, and quasi-experimental program expansions. Lottery evidence shows that students least likely to apply benefit most from choice programs. To evaluate alternative designs, we estimate a generalized Roy model of applications, enrollment, and achievement. Specialized choice schools raise achievement by 0.25 standard deviations, but application costs screen out high-return students and leave seats unused. Counterfactuals show mandatory participation with deferred acceptance reduces these inefficiencies, generating larger academic gains compared to information or transportation policies. System design, not school effectiveness alone, determines who benefits from public school choice.

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We are grateful for insightful comments from Marianne Bertrand, John Friedman, Nathan Hendren, Matt Notowidigdo, Minseon Park, Canice Prendergast, Chris Walters, and seminar participants at Arizona State University, Opportunity Insights, Rochester University, University of Michigan, University of Chicago, and University of Maryland. Ryan Lee and Connor Fogal provided outstanding research assistance. Campos acknowledges support from the John E. Jeuck Faculty Fellowship.

KEYWORDS: school choice, market design, centralized assignment, magnet schools, selection on gains, educational inequality.
JEL CLASSIFICATION: I21, I24, J24.

1. INTRODUCTION

Over the last two decades, U.S. public school districts have aggressively expanded school choice—opening new magnet schools, dual-language programs, theme-based academies, and selective-admissions exam schools that collectively provide new intra-district options outside of the traditional neighborhood system. These efforts were motivated by a desire to reduce persistent inequities in educational opportunity and by competitive pressures from charter schools (Kahlenberg et al., 2019, Hess et al., 2001). The need for such reforms is stark: within the same district, students who attend the lowest-performing schools fall behind their peers at the highest-performing schools by roughly 0.20 standard deviations in achievement per year, a gap that education researchers equate to about two-thirds of a grade level of annual learning.¹ Yet, grafting choice onto a framework built on residential assignment creates an immediate market design challenge: how should these new education markets be organized so that access is equitable and seats are efficiently allocated? The potential benefits of school choice—through competition and better student-school matches (Campos and Kearns, 2024, Ovidi, 2025, Bruhn et al., 2023, Hoxby, 2003)—will materialize only if families, especially those who stand to gain the most, are able to navigate these options and submit informed applications.

This paper provides new evidence on the structure of public school choice systems and how their design shapes educational opportunity. To motivate our analysis, we begin by documenting the national landscape of school choice—and how districts have tackled the market design problem—based on original data collected from the nation’s 150 largest school districts. Enrollment in non-neighborhood public schools of choice has more than doubled over the past two decades, with enrollment shares now exceeding those of the charter sector. We also uncover substantial heterogeneity in how families are incentivized to participate. Most large districts (64%) have adopted algorithms to assign students to schools, but a majority also require parents to proactively “opt-in” and submit applications.

¹We calculate this statistic by averaging the 90-10 differential in within-district test score growth across all U.S. districts that have at least 10 schools reported in the Stanford Education Data Archive (SEDA). SEDA has harmonized test scores to be comparable across states and districts.

By contrast, only 9% of districts require all students to submit preferences by mandate, as is done in cities like New York and Boston ([Abdulkadiroğlu and Sönmez, 2003](#), [Abdulkadiroğlu et al., 2017](#)). Districts with opt-in participation also provide less clear information and have more complex applications, creating frictions that raise the cost of entry. Consistent with this, we document marked under-representation of economically disadvantaged students in choice schools. These patterns suggest that voluntary participation may segment the public schooling by allowing the most advantaged students to leave behind their less advantaged peers.

Motivated by these findings, we study the consequences of design: how do the institutional features of opt-in systems shape achievement and inequality, and more generally, how would other designs perform? Following [Roth \(2015\)](#), we organize our analysis around the questions of who participates, what benefits choice schools provide, and how market design shapes the allocation of those benefits. The first concerns selection into participation: which students opt in and which do not? The second concerns whether choice schools are vertically differentiated and generate meaningful achievement benefits for those who enroll. The third concerns how institutional features, including participation rules (e.g. [Ricks, 2025](#)) and the information environment (e.g. [Neal and Root, 2024](#)), shape access to those schools and the resulting outcomes.

To bring these questions into focus, we turn to the Los Angeles Unified School District (LAUSD), the largest voluntary public choice system in the U.S. and a rich environment for analyzing how design shapes participation and outcomes. Treated as its own district, LAUSD's choice program would have the 23rd-highest enrollment in the country, serving roughly 110,000 students annually. Notably, the number of choice options has more than doubled in the past two decades, reflecting both parental demand and competitive pressures from the rapidly expanding charter sector. LAUSD's choice school portfolio now includes hundreds of magnet programs, fifty affiliated charters, and an expanding set of dual-language and theme-based schools. Our analysis relies on nearly two decades worth of detailed application records, linked to district-wide administrative student data containing achievement and other key outcomes. Importantly, the data also include information on randomized admission lotteries conducted at oversubscribed choice programs. We focus on middle-school applications, where achievement impacts can be observed soon after enrollment.

Our analysis of who participates in LAUSD’s opt-in choice system combines two exercises: a descriptive characterization of applicants and a quasi-experimental analysis of how participation responds to the expansion of programs. Choice schools in LAUSD’s opt-in system attract a student body that is more racially diverse while also being more economically and academically advantaged. Applicants have much stronger academic preparation—the baseline achievement is roughly 0.8σ higher than non-applicants—and are less likely to be low-income or English-learners. Proximity also plays a central role—students who live closer to a choice program are much more likely to apply. To understand the importance of distance better, we leverage the unprecedented expansion of LAUSD’s choice sector, which generated quasi-random changes in access. Event study estimates show that neighborhoods experiencing reductions in relative distance saw participation rates rise by 17% on average. These findings demonstrate that while academically advantaged students tend to sort into choice programs, supply-side policy can broaden access through program expansion or reductions in travel costs. Whether that expanded participation translates into achievement gains and reduced inequality, however, depends on which students stand to gain the most from attending a choice program.

Next, we turn to nearly two decades of lottery data to examine which students gain the most from opting-in. Motivated by our analysis of application behavior, we estimate how treatment effects vary with distance to a choice program. We find that students who live close to choice programs benefit the most from attendance: The difference in reduced form impacts between students in the first and the fifth quintile of distance to choice school is roughly 0.1σ . This finding suggests that students with the lowest *unobserved* demand to “opt-in” also stand to gain the most from attendance. We further probe this interpretation with two supplemental exercises. The first leverages the fact that application rates vary greatly by student subgroups (e.g., race or baseline ability) to ask whether students with low *observable* demand also benefit more. The second leverages plausible preferences for homophily (as in [Corradini and Idoux, 2025](#)) to further probe the role of *unobservable* determinants of demand in a manner that is similar in spirit to the distance-based patterns. In both cases, we find similar results: low-demand predicts a larger benefit. Thus, taken together, our results to this point suggest that opt-in designs generate allocative inefficiency by screening out high-return students.

1 A key policy implication of this reduced form evidence is that the optional nature of 1
2 participation can limit the benefits of public school choice. To evaluate whether system- 2
3 level reforms could improve outcomes, we develop and estimate a structural model linking 3
4 school applications, enrollment, and achievement. The model recovers the distribution of 4
5 treatment effects for all students and not just the set of students who currently select into 5
6 choice programs through the opt-in system. We use a generalized Roy model of application 6
7 and enrollment demand that follows [Walters \(2018\)](#) by linking idiosyncratic school prefer- 7
8 ences to potential outcomes. The link provides control functions for selection and param- 8
9 eterizes treatment effect heterogeneity. For identification, we use two sources of variation: 9
10 randomized admission offers at oversubscribed programs and policy-driven expansions of 10
11 LAUSD choice schools that altered students' relative distance. While lottery offers are 11
12 standard for identification in this literature, there has been concern over the use of cross- 12
13 sectional instruments based on distance (e.g. [Carneiro and Heckman, 2002](#)). Rather than 13
14 relying on cross-sectional differences in student proximity, we therefore exploit the quasi- 14
15 experimental changes induced by the choice sector's expansion, which we show balance 15
16 well on baseline attributes. Together, these features provide plausibly exogenous variation 16
17 that allows us to obtain credible estimates of causal effects, which feed into our final coun- 17
18 terfactual analysis of how various alternative policies shape enrollment patterns and student 18
19 achievement. 19

20 Our selection-corrected estimates show that choice schools are higher quality, generating 20
21 average treatment effects of 0.25σ for math and 0.23σ for English Language Arts (ELA) 21
22 achievement. These findings from LAUSD provide evidence that districts facing competi- 22
23 tive pressures, such as from charter schools, respond by creating a vertically differentiated 23
24 product and expanding it. At first glance, the combination of vertical differentiation and 24
25 strong selection based on prior achievement into choice schools through an opt-in applica- 25
26 tion design might exacerbate educational inequality. However, we find a negative associa- 26
27 tion between preferences and causal effects, indicating that those most likely to opt in also 27
28 gain the least—patterns consistent with our earlier reduced-form analysis. The difference 28
29 in average effects between students with the weakest and strongest proclivity for choice 29
30 school enrollment is on the order of magnitude of -0.30σ , indicating that selection into 30
31 treatment based on gains dilutes the impacts of opt-in systems on inequality in achieve- 31

ment outcomes. Overall, the characterization of treatment effects for the entire population demonstrates that there is scope for improvements in district-level achievement.

With the structural estimates in hand, we turn to our counterfactual analysis exploring how alternative policies affect sorting and academic performance relative to a baseline simulation of LAUSD’s current opt-in system. Throughout the counterfactuals, we hold fixed the existing menu of programs and school capacities. Our analysis starts by examining participation-focused policies that lower key access barriers. These include simulations of (i) information interventions that lower search frictions (Deming et al., 2014, Corcoran et al., 2018, Hastings and Weinstein, 2008, Ainsworth et al., 2023, Neal and Root, 2024), and (ii) busing-type policies that reduce travel costs (Setren, 2024, Angrist et al., 2022). We then turn to studying overall system design features by simulating outcomes under (iii) a decentralized market (in which families can apply to multiple schools and receive multiple offers) and (iv) centralized assignment with mandatory participation and deferred acceptance (Lincove and Valant, 2023). Finally, we combine participation-enhancing policies with a mandatory assignment regime to evaluate how these reforms jointly shape sorting and achievement. In theory, these counterfactual policies can expand access by lowering information barriers or application costs and encourage participation among students who stand to gain the most. At the same time, each reform also introduces trade-offs as broader participation changes who enrolls and can create capacity pressures, motivating formal counterfactual analysis of how assignment design reshapes access, sorting, and achievement.

We find that general or targeted information interventions modestly broaden participation, while mandates mechanically increase participation. As participation widens, average achievement rises, with the largest gains under a fully centralized system (i.e., mandating participation and using a central clearinghouse for assignment). By contrast, a decentralized market results in more advantaged students occupying more seats in vertically differentiated schools and in modest gains in district-level achievement relative to the opt-in status quo. Pairing reduced travel costs (via a busing-like policy) with centralization produces modest additional gains, highlighting that high application costs in an opt-in system are more of a barrier than travel costs. In terms of winners and losers of the fully centralized system, we find that all groups experience improvements in average outcomes. These results reflect the elimination of two distinct inefficiencies present in an opt-in system. The

first is negative sorting on gains, whereby students with substantial gains to participating self-select out of participation. The second is substantial slack in the system, as many seats at high-quality schools go unfilled. Centralizing participation mostly eliminates both forms of inefficiency, producing achievement gains of 0.018σ for math and 0.016σ for ELA. The impacts on students who actually enrolled in choice schools (i.e., the treatment on the treated impacts) amount to $0.18 - 0.21\sigma$, compared to $0.08 - 0.11\sigma$ in the status quo opt-in allocation.

To further characterize the potential gains from our counterfactual policies, we conduct a simulation that removes all frictions and allocative inefficiencies by directly matching students to schools to maximize district-level achievement.² Relative to this achievement-optimal benchmark, we find that centralization captures nearly 50% of the potential district-level gains in math and ELA, while combining centralization with busing captures over 55 percent. Although none of the policies we study reach the achievement-optimal allocation, the remaining gap points to a fundamental limitation: even when participation is universal and capacity is fully used, demand for effective schools remains imperfectly aligned with school quality. We demonstrate that this misalignment arises because families' preferences reflect a mix of academic, social, and informational factors, leading many to favor schools that are not the most effective. Consequently, while centralization substantially improves the efficiency of school assignment, the overall gains from reform are bounded by how demand for school effectiveness shapes equilibrium outcomes.

Ultimately, our analysis yields two main policy lessons. First, rapidly expanding district-run choice programs—often introduced in response to charter school competition—are vertically differentiated and, on average, raise student achievement, indicating that districts respond competitively when the K-12 landscape shifts. Second, layering voluntary choice on top of neighborhood assignment poses a market-design challenge, and the approach most districts now use appears to deepen inequality in access to opportunity: negative sorting on gains produces allocative inefficiencies, while complex applications and limited outreach leave many high-quality seats empty. Thus, expanding families' options marks real

²Following [Abdulkadiroğlu et al. \(2025\)](#), this treatment effect maximizing allocation solves a linear program that maximizes student achievement subject to capacity constraints.

progress, yet its promise will be met only when assignment mechanisms are tuned to dis-
tribute those opportunities equitably—the devil is in the details.

2. RELATED LITERATURE

The main contribution of this paper is to the literature on education market design and student outcomes. Prior work shows that coordinated assignment can reduce mismatch and improve welfare (Abdulkadiroğlu et al., 2017, Prendergast, 2017), while related work studies information provision and mechanism design within centralized systems (Calsamiglia et al., 2020, Corcoran et al., 2018, Corradini, 2023, Ainsworth et al., 2023, Kapor et al., 2020, Agarwal and Somaini, 2018). We shift attention to a less-studied margin: participation in public school choice itself. Using new national data, we document that opt-in systems are the modal design and, building on models of self-selection (Walters, 2018), show how participation rules can shape sorting, achievement, and allocative efficiency.

We also relate to literature showing that school choice can increase sorting and stratification across schools (Urquiola, 2005, Hsieh and Urquiola, 2006, Munteanu, 2024, Machado and Szerman, 2021) and that preferences and travel costs can limit access (Laverde, 2024, Idoux, 2022). Our contribution is to show that the participation rule itself is a first-order determinant of who enters choice markets and of the resulting academic outcomes.

Finally, our findings connect to the literature on how public schools respond to competition (Hoxby, 2000, 2003, Figlio and Hart, 2014, Gilraine et al., 2021, Campos and Kearns, 2024, Crema, 2022). Charter schools are vertically differentiated and can draw enrollment from traditional districts (Abdulkadiroğlu et al., 2011, Dobbie and Fryer Jr, 2011, Angrist et al., 2013, Mumma, 2022). We show that district-run public school choice has expanded alongside charter competition and is itself vertically differentiated. Further, our results show that the effectiveness of this competitive response depends on how districts structure participation and assignment.

3. PUBLIC SCHOOL CHOICE SYSTEMS ACROSS THE U.S.

While charter schools have attracted substantial media coverage and policymaker attention in recent years, an equally important transformation has taken place within traditional public school districts themselves. Many large districts now operate extensive “intra-district” choice systems that allow families to select among magnet, dual-language, and

1 themed programs outside their neighborhood zones. Panel (a) of Appendix Figure [A.1](#) 1
2 shows that, among the 150 largest school districts, enrollment in district-run choice schools 2
3 has more than doubled since 1999 and consistently exceeded charter school enrollment. 3
4 More than 15% of all students are now educated in a district-run choice option, i.e., over 4
5 40% larger than the corresponding market share going to local charters. 5

6 The consequences of this expansion depend on how districts organize participation and 6
7 assignment. Systems differ in whether participation is mandatory or voluntary, and whether 7
8 assignments are centralized. For example, Boston Public Schools, requires that all families 8
9 rank schools and submit them to a centralized clearinghouse, which then determines assign- 9
10 ments using an algorithm ([Abdulkadiroglu et al., 2006](#)). In contrast, Miami-Dade County 10
11 Public Schools defaults families to neighborhood schools unless they voluntarily opt into 11
12 a centralized system that determines assignment. Districts also vary in the degree to which 12
13 families are able or willing to participate.³ These examples illustrate the diversity of local 13
14 policies, while also underscoring the limits of what we know: they come from just a handful 14
15 of districts. To date, there has been no systematic effort to document how choice systems 15
16 are actually structured across the U.S., despite the fact that these design features are central 16
17 to understanding welfare effects. 17

18 To learn about the structure of public school choice systems broadly, we collected orig- 18
19 inal data for the 150 largest U.S. school districts. We used a hybrid approach: key infor- 19
20 mation was obtained through direct phone interviews with district representatives and sys- 20
21 tematic reviews of district websites. We combined this original data with publicly available 21
22 data on enrollment shares from the Common Core, producing data covering districts edu- 22
23 cating roughly 27% of all U.S. public school students. Appendix [B.1](#) provides additional 23
24 data collection details and summary statistics. 24

25 Figure 1 summarizes our three key findings. First, mandatory assignment, the design 25
26 emphasized in much of the school-choice literature, is rare. Only 8.8% of districts use 26
27 mandatory participation with centralized assignment. By contrast, 55.1% operate opt-in 27
28 districts centralized systems, 34.7% use decentralized applications an 1.4% offer no sys- 28

30 ³One barrier is that parents may lack information and misunderstand features of their local system. For example, 30
31 reporting on San Francisco's lottery system highlights how parents perceive the process as confusing and stressful, 31
32 emphasizing the gap between the system's intended design and how it is understood by families ([Li, 2025](#)). 32

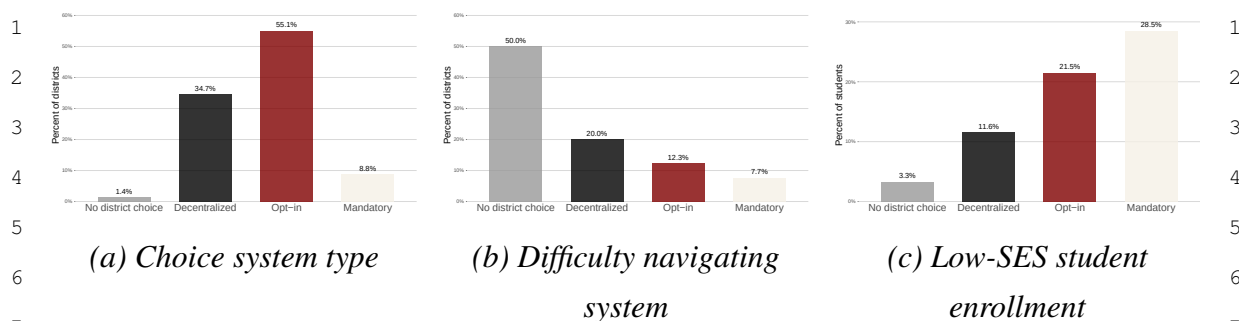


FIGURE 1.—Different Systems of Choice Among the 150 Largest U.S. School Districts. *Note:* This figure reports statistics from our original data collection. Panel (a) classifies districts as having no district choice, decentralized applications without a clearinghouse, an opt-in centralized system, or mandatory preference submission with centralized assignment. Panel (b) reports the share of each type that is difficult to navigate; for districts with no choice, this means difficulty determining whether intradistrict choice exists. Panel (c) reports choice-program enrollment among students eligible for free or reduced-price lunch (FRPL). We omit six districts with FRPL rates below 10% because universal meal programs make their reporting incomplete.

tematic intra-district choice.⁴ In sum, voluntary participation with centralized assignment is the modal design among large U.S. districts.

Second, navigation difficulty declines as choice becomes more centralized. We classify a district as difficult to navigate if both the website and phone pathways required multiple steps to learn whether choice exists and how to apply. Half of districts without systematic choice and 20% of decentralized districts were difficult to navigate, compared with 12.3% of opt-in centralized districts and roughly 7% of mandatory centralized districts. These patterns suggest that institutional design is closely related to the practical costs families face when accessing school choice.

Third, more centralized systems enroll a larger share of low-SES students in choice schools. Choice enrollment among low-SES students averages 3.3% in districts without a systematic choice and 11.6% in decentralized systems, compared with 21.5% in opt-in centralized systems and nearly 30% in mandatory systems with centralized algorithms. Taken together, our findings on the inverse relationship between navigation difficulty and low-SES participation motivates our central question: do the students least likely to participate in opt-in systems also stand to gain more from access to choice schools? In the next

⁴We do not classify hardship or capacity-contingent boundary waivers as intra-district school choice.

section, we investigate this question by conducting a detailed analysis of the largest opt-in district in the county, LAUSD.

4. LAUSD: BACKGROUND AND DATA

LAUSD's opt-in system developed on top of a traditional neighborhood-assignment structure. Since 1982, the district has layered an expanding set of choice options onto attendance-zone assignment. A court-ordered desegregation mandate spurred the growth of magnet programs, followed by charter schools in the 1990s. In the 2000s, additional initiatives culminated in the Zones of Choice program for high-school students, which created controlled choice environment in a subset of neighborhoods.

Since the late 2000s, LAUSD has aggressively expanded its portfolio of choice schools. This reflects competitive pressures from the charter sector and declining district enrollment. As a result, the district more than doubled the number of magnet programs, introduced dozens of affiliated (i.e., district-run) charter conversions, and scaled up dual-language and theme-based offerings. Panel (b) of Appendix Figure A.1 documents this growth between 2004 and 2023.⁵

Many choice programs are oversubscribed and use randomized admission lotteries. Yet, despite oversubscription at particular campuses, there is nontrivial slack in the system because capacity growth has outpaced demand in recent years. In the period most relevant for our analysis (2004-2013), aggregate capacity utilization was around 60%. This coexistence of oversubscription and system-wide slack is central to our counterfactual analysis, which holds system-wide capacity constant and examines how alternative design features reallocate students across programs.

LAUSD's School Assignment Policies As highlighted, LAUSD is the largest opt-in district nationally. From their inception under the district's court ordered desegregation plan, LAUSD magnet schools have allocated seats using a centralized lottery system. Unlike zoned schools, magnets accept students citywide, so the district created a central application and lottery process. Because magnets and choice schools are not tied to neighborhood attendance boundaries, citywide eligibility can imply substantial cross-neighborhood

⁵Supplemental Appendix A.1 provides additional detail on LAUSD's major choice program types.

commuting. Consistent with this, LAUSD provides district-paid transportation for some students beyond grade-specific distance thresholds (e.g., secondary school students are eligible if they live outside a 3-mile radius). Early on, magnets included racial enrollment quotas in their admission process to fulfill their integration mission. This use of race in assignments continued after court supervision ended and was later upheld as lawful despite California’s Proposition 209 ban on racial preferences ([American Civil Rights Foundation v. Los Angeles Unified School District, 2008](#)).

Historically, LAUSD’s opt-in assignment system allowed families to rank one option, a single-application feature that is central to our empirical analysis. More recently, families have been able to submit rank-ordered lists for magnet and dual-language programs through a Unified Enrollment platform, while district-run charters continue to run their own applications systems. Oversubscribed programs use lottery numbers to break ties, with recent assignments following a variant of Immediate Acceptance. Participation frictions commonly emphasized in the K-12 market design discourse (e.g., search and information frictions, coordination costs, and the implications of strategic complexity) are also present in LAUSD. However, the opt-in requirement itself may be a much larger participation friction, particularly in a relatively weak information environments (see Panel (b) of Figure 1).

Data and Samples Our analysis uses LAUSD administrative data for fifth-grade cohorts from 2004-2017, whom we follow through Grades 6–8. The data include demographics, baseline standardized test scores in Math and ELA, residential Census block, and choice school applications. Our primary achievement outcomes are student-level averages of available Math and ELA test scores.⁶

We use three primary analysis samples. The baseline sample includes all fifth-grade cohorts from 2004 and 2017. The lottery sample contains students who applied to an oversubscribed program and participated in one of 1,033 lotteries; we further require valid residential geocodes to construct school-distance measures. Lotteries are conducted within strata defined by the cross of application year, student grade, program, race, and priority-points

⁶Achievement outcomes are observed only for students who remain enrolled in LAUSD. Section 6 shows that our reduced form results are robust to imputing missing post-lottery test scores using prior year test scores.

level.⁷ Finally, the structural sample is a subset of the baseline sample, restricted to students who (i) have valid geospatial data, (ii) enroll in an LAUSD school in sixth grade, (iii) have at least one test score in Grades 6–8, and (iv) belong to the 2004–2013 cohorts. This final condition restricts estimation to student cohorts who could apply to at most one school, which allows our analysis to sidestep common issues associated with strategic play or the curse of dimensionality (Agarwal and Somaini, 2018).

Appendix Table A.1 provides summary statistics comparing the three samples. Lottery applicants are positively selected, with stronger prior achievement, roughly 0.65σ above district averages in both Math and ELA. By contrast, the structural sample closely resembles the baseline population, aside from the fact that it includes more English learners and slightly lower fifth-grade achievement.

5. WHO PARTICIPATES IN CHOICE SCHOOLS?

We begin by studying selection into LAUSD’s choice schools during 2004–2017. Table I presents fifth-grader descriptive statistics comparing non-applicants (column 1), applicants (column 2), and the estimated differences between the two groups (column 3). We find that participation in the school choice system is limited: only 20% of students submit an application.

Applicants differ systematically from the broader LAUSD student population along demographic and socioeconomic lines. Hispanic students constitute 79% of non-applicants but only 54% of applicants. In contrast, Black, White, and Asian student shares are over-represented among applicants by 4, 12, and 8 percentage points, respectively. Applicants are also socioeconomically advantaged: poverty and English Learner rates are 15 and 21 percentage points lower relative to non-applicants. Female students are slightly overrepresented, comprising 51% of applicants compared with 49% of non-applicants.

Choice applicants are substantially higher achieving, with baseline Math and ELA scores nearly 0.8σ above non-applicants. This positive selection on prior achievement is consistent

⁷Supplemental Appendix B details the lottery data and our procedure for inferring each lottery’s cutoff for extending offers. We construct an indicator equal to one if an applicant’s random number lies on the offer side of the inferred cutoff; this is our measure of receiving a lottery-based choice school offer.

TABLE I
SUMMARY STATISTICS: LAUSD GRADE 5 STUDENTS (2004-2017)

	Non- applicants (1)	Applicants (2)	Diff. (s.e.) (3)
<i>Student demographics</i>			
Hispanic	0.79	0.54	−0.25*** (0.02)
Black	0.09	0.13	0.04*** (0.01)
White	0.07	0.19	0.12*** (0.01)
Asian	0.03	0.11	0.08*** (0.01)
Female	0.49	0.51	0.02*** (0.00)
Poverty	0.75	0.60	−0.14*** (0.01)
English learner	0.31	0.10	−0.21*** (0.01)
<i>Standardized test scores</i>			
ELA	−0.18	0.63	0.81*** (0.03)
Math	−0.16	0.60	0.77*** (0.03)
<i>Distance to district schools</i>			
Nearest choice (relative distance)	0.59	0.51	−0.08*** (0.02)

Note: Summary statistics for LAUSD fifth-grade cohorts from 2004–2017. Columns 1 and 2 describe non-applicants and applicants, respectively; Column 3 reports mean differences, with standard errors in parentheses. Relative distance is the distance to the nearest district choice school minus the distance to the nearest district non-choice school.

with evidence from studies of charter and highly selective schools (Deming et al., 2014, Abdulkadiroğlu et al., 2011).

Finally, proximity to schools is closely connected to participation. Appendix Figure A.2 shows that application rates fall from just above 24% among students living closest to a choice school to about 18% among those living furthest away. This pattern is consistent with prior research documenting that distance is a critical factor in the demand for high-quality schooling options (Laverde, 2024, Hastings et al., 2009).

5.1. Does Distance Determine Participation? Evidence from Choice Expansion

To more clearly test the importance of distance as a shifter of demand, we use LAUSD's rapid expansion of choice programs as an empirical testing ground. This analysis addresses the concern that residential proximity may be correlated with parental motivation, information, or neighborhood characteristics. Our analysis exploits the fact that the district more

than doubled the number of choice programs over the past two decades, generating substantial within-neighborhood variation in proximity. Appendix Figure A.3 illustrates the resulting changes in geographic access between 2004 and 2023.

We estimate an event study model that compares changes in application rates across neighborhoods (tracts) that did and did not experience a reduction in travel distance due to choice program expansion. Treated tracts are those that experience a reduction in distance to the nearest choice school of more than half a mile in a given year. Control tracts are those that saw no change in distance during the previous three academic years. To mitigate spatial spillovers, we exclude potential control tracts located within 2.5 miles of a treated tract’s centroid. We organize this sample of tracts into a stacked event-study sample, where each “event” corresponds to the first year a tract experiences a qualifying distance reduction.⁸

Our formal analysis uses a stacked sample and estimates:

$$Y_{nts} = \alpha_{ns} + \alpha_{ts} + \sum_{k \neq -1} \beta_k D_{ntk} + u_{nts}, \quad (1)$$

where Y_{nts} is an outcome for neighborhood n in event-time t and stack s , α_{ns} are neighborhood-by-stack fixed effects and α_{ts} are event-time-by-stack fixed effects, and $D_{ntk} \equiv \mathbf{1}\{\text{event-time} = k\} \times \mathbf{1}\{\text{Neighborhood } n \text{ is treated}\}$, and u_{nts} is an error term. The inclusion of stack-specific fixed effects restricts to within-stack comparisons of changes in outcomes between treated and control units. The coefficients β_k are average differences in changes in outcomes between treated and control units between time k and the year before treatment ($k = -1$).

Figure 2 shows a sharp increase in applications at program entry. Treated neighborhoods see a roughly 20% increase in neighborhood application rates. Pre-trends are flat, providing reassurance for the parallel-trends assumption underlying our design. Together with the evidence in Table I and Appendix Figure A.2, these quasi-experimental results show that changes in distance causally affect choice-school demand in LAUSD.

⁸In the stacked sample, we re-index time relative to that event year, pooling across all cohorts to trace out dynamic responses to improved access.

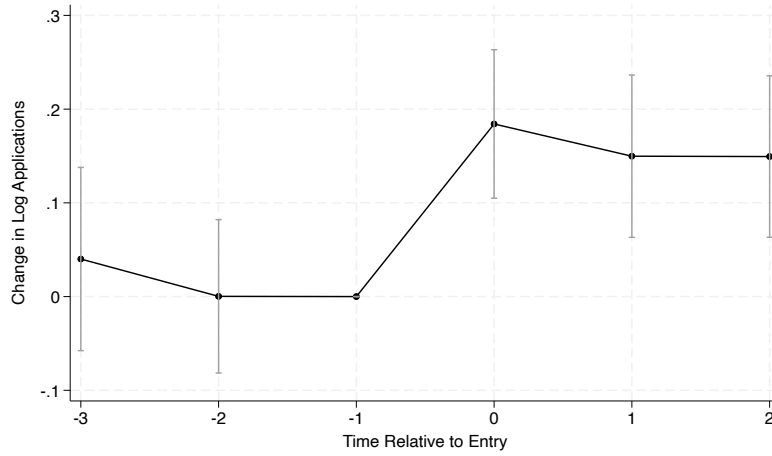


FIGURE 2.—Changes in Application Rates in Response to Choice School Entry. *Notes:* This figure reports event-study estimates based on Equation (1); the dependent variable is the log of neighborhood applications to choice schools. An event is the first academic year in which a census tract experiences a reduction of more than 0.5 miles in its distance to the nearest choice school. Control tracts exhibit no change in distance during the prior three academic years and lie at least 2.5 miles from the centroid of every treated tract. The sample includes only tracts with nonzero LAUSD enrollment. Bars report 95% confidence intervals using robust standard errors clustered by neighborhood.

6. IMPACTS OF CHOICE SCHOOLS: LOTTERY-BASED EVIDENCE

Having established who participates in LAUSD's opt-in choice system, we next study which students gain the most from choice program enrollment. We use a flexible approach to estimating causal effects of choice schools that exploits oversubscribed program lotteries which allocate seats using random priority number lotteries within strata defined by program, year, grade, priority points, and race combinations.⁹

Motivated by the relationship between application behavior and proximity in the preceding section, we estimate how the reduced form impacts of an offer varies with baseline distance to the nearest choice school. We estimate how the ITT effect varies using the following specification:

$$Y_i = \alpha_{\ell(i)} + \sum_{q=1}^5 \beta_q (Z_i \times \mathbf{1}(Q_i = q)) + \sum_{q=1}^5 \kappa_q \mathbf{1}(Q_i = q) + u_i, \quad (2)$$

⁹Many programs use race-specific admission strata as part of LAUSD's longstanding desegregation framework.

where Y_i is a post-lottery achievement outcome for student i , Q_i denotes their baseline distance to the nearest choice school quintile, $\alpha_{\ell(i)}$ are lottery fixed effects, and Z_i is an indicator for whether applicant i received a lottery offer. Because Z_i is randomized within lottery strata, β_q identify ITT effects *within* distance bins. Appendix Table A.2 shows covariate balance within each distance quintile, while Appendix Table A.3 shows that the results are unchanged when we impute missing post-lottery test scores using pre-lottery baseline scores.¹⁰ Standard errors are clustered at the lottery level.

Figure 3 reports our main reduced-form estimates.¹¹ Math effects decline sharply with distance: students in the nearest quintile gain roughly 0.05σ from offers, while those in the most distant quintile lose about 0.05σ , and we reject equality across quintiles. ELA exhibits a qualitatively similar gradient. First-stage effects are comparable across distance quintiles (0.38-0.45; Appendix Table A.4), so differential take-up does not explain the pattern. Because the first and fifth quintile effects are opposite in sign, IV rescaling will mechanically magnify the gradient in impact between the first and fifth quintiles.

How should we interpret this pattern of results? The distance-based gradient in treatment effects is consistent with findings from Walters (2018), who documents similar heterogeneity in charter-school lotteries: students who live farther from charter schools—those facing higher travel costs and thus revealing stronger demand—tend to experience smaller achievement gains once enrolled. This pattern implies a form of reverse Roy sorting, in which students with the greatest willingness to participate are not those who benefit the most.

We conduct two exercises to probe the reverse-Roy interpretation. First, we ask whether observable heterogeneity provides an alternative explanation for the reverse-Roy pattern or whether differences in outside options can account for the distance gradient. Appendix Figure A.6 compares treatment effects across student subgroups with different application rates. Each set of point estimates corresponds to a subgroup and its complement (e.g., low-

¹⁰Because achievement is observed only for students who remain in LAUSD, differential attrition could in principle bias the estimates. The balance and imputation exercises provide little evidence that this drives our results.

¹¹This heterogeneity is obscured in a pooled specification that ignores distance; the average ITTs are -0.027σ in math ($p < 0.01$) and -0.036σ in ELA ($p < 0.01$).

versus high-achiever), so we can compare within each pair. In nearly all cases, the group with the higher demand has the smaller estimated treatment effect, further evidence of reverse-Roy sorting. A Sargan-Hansen test also rejects the hypothesis that subgroup differences in take-up alone explain the corresponding reduced-form heterogeneity in both Math and ELA ($p < 0.001$).¹² Appendix Figure A.4 then asks whether the distance gradient is explained by observable treatment-effect heterogeneity or differences in fallback-school quality. Allowing offer effects to vary with baseline achievement, poverty, other demographics, and neighborhood-school value added leaves the gradient essentially unchanged. Thus, neither observable student heterogeneity nor differences in outside options account for the pattern.¹³

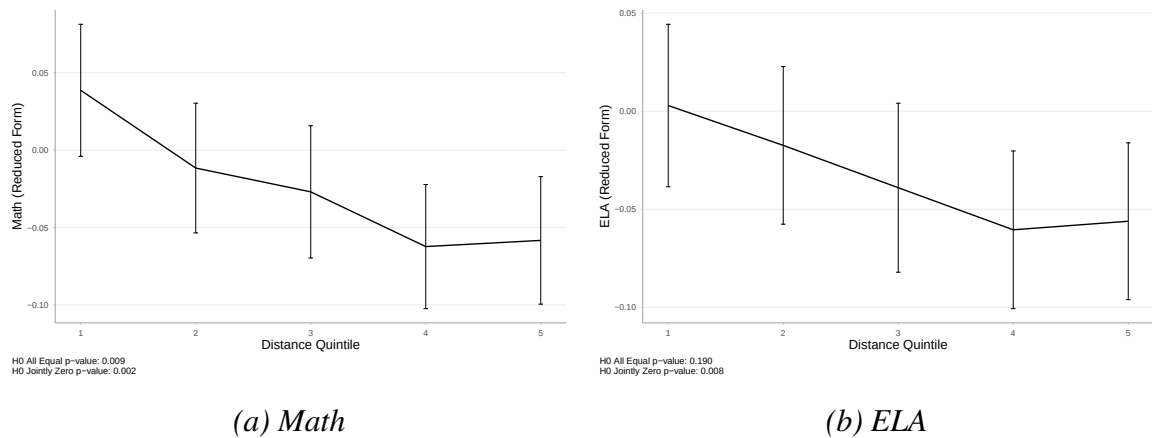


FIGURE 3.—Oversubscribed Choice School Lottery Effects by Distance Quintile. *Notes:* This figure reports reduced-form effects of receiving an offer on Math and ELA achievement in Panels (a) and (b), respectively, by quintile of baseline distance to a choice school. Estimates are from regressions of achievement on lottery-stratum fixed effects, distance-quintile main effects, and interactions between offers and distance quintiles. Bars report 95% confidence intervals using robust standard errors clustered at the lottery-stratum level. The figure also reports p -values for equality of the distance-specific effects and for their joint equality to zero.

¹²The test asks whether heterogeneity in subgroup-specific first stages is sufficient to explain heterogeneity in reduced-form effects. Given randomized lottery offers, rejection points to additional treatment-effect heterogeneity.

¹³A separate concern is that commuting distance directly affects achievement. Prior evidence generally finds negligible test-score effects of longer travel (e.g., Angrist et al., 2022); near the end of Section 8, we address this possibility directly in the structural model.

Second, we ask whether the same pattern appears using an alternative proxy for latent demand. Prior research documents that parents and students prefer to have peers who are observably similar to themselves (Hastings et al., 2009, Corradini and Idoux, 2025). Under such preferences, applicants who are less similar to the typical student to their program must have stronger idiosyncratic demand to offset that mismatch. In line with this, we construct a peer-similarity index based on the Euclidean distance between each applicant’s observable characteristics and those of the average applicant to a given program. Lottery effects by quintiles of this index (Appendix Figure A.5) show a nearly identical negative gradient: students least similar to their peers—those likely to have the strongest unobserved demand for the program—experience the smallest gains. This complementary evidence reinforces the interpretation that unobserved demand is an important source of treatment effect heterogeneity.

7. BEYOND LOTTERY EFFECTS

The lottery estimates provide evidence on the effects of choice programs for families who apply. To understand how the opt-in design shapes district-wide outcomes and inequality, we require a framework that connects application and enrollment decisions to potential outcomes for the broader population of students. This section develops such a model in the spirit of Walters (2018) and Avery et al. (2025).

7.1. Setup and Timing

Our framework models the optimal choice problem faced by forward-looking students and their families when deciding whether to apply and which choice program to select. Supplemental Appendix Figure C.1 illustrates the sequence of decisions and events. In Stage 1, consistent with our interest in LAUSD and its policies during the 2004–2013 academic years, each rising sixth-grade student i chooses whether to apply to a single choice school program $j \in \{1, \dots, J\}$, with the binary variable A_{ij} indicating an application to school j . Students are always guaranteed access to their neighborhood school, denoted $j = 0$. In Stage 2, each choice school $j > 0$ makes admission offers to applicants, with Z_{ij} denoting whether the student receives an offer; when schools are oversubscribed, admission is determined by randomized lotteries, generating exogenous variation in offers. In Stage 3, students make enrollment decisions over programs, denoted by $E_i \in \{0, 1, \dots, J\}$

where $E_i = 0$ indicates neighborhood school attendance. Finally, students attend their chosen schools, take achievement exams, and realize academic outcomes Y_i in Stage 4.

7.2. Student Preferences

Let $i \in \mathcal{I}$ index rising sixth-grade students in LAUSD and $j \in \mathcal{J}$ index the set of choice schools. Each student may apply to any choice school and, if admitted, enroll there; otherwise, they default to their neighborhood school ($j = 0$) which varies across neighborhoods.¹⁴ We define the indirect utility of student i from enrolling in choice school j relative to their neighborhood school as: $U_{ij}^* = U_{ij} - U_{i0}$, so that $U_{ij}^* > 0$ indicates a strict preference for choice school j over the default.

This utility difference is specified as follows:

$$U_{ij}^* = \underbrace{\theta_i + \beta_c X_i + \delta_j - (\psi_D D_{ij} + \psi_{D2} D_{ij}^2 + \psi_{Dx} D_{ij} \times X_i)}_{v_j(\theta_i, X_i, D_i)} + \epsilon_{ij}, \quad (3)$$

where X_i contains student-level characteristics, and D_{ij} is relative distance, defined as the difference between student i 's distance to school j and their nearest non-choice school. The term δ_j measures the mean popularity of choice school j and absorbs average differences in school quality and preferences over achievement. This model allows for two main sources of preference heterogeneity for each student i . The first is observable preference heterogeneity embedded in the vector X_i , which includes race, socioeconomic status, baseline achievement, neighborhood quality, and baseline peer achievement scores among other common demographic characteristics. The preference vector β_c does not depend on j , implying that observable preference heterogeneity applies uniformly across all choice schools. Second, the term θ_i allows for unobservable preference heterogeneity. We model travel costs of distance flexibly by specifying a quadratic function for D_{ij} that also features an interaction between distance and the observable student characteristics X_i . Finally, ϵ_{ij} represents idiosyncratic heterogeneity in student tastes for specific school-student pairings, which also encompasses school-specific match effects in realized achievement.

¹⁴We take the assigned neighborhood school as the outside option because our object of interest is the gain from accessing the district choice system rather than the full distribution of public school effects.

The individual taste parameter θ_i is important in our model of student preferences. A few points about θ_i are worth highlighting. First, because θ_i enters identically for every $j > 0$, it shifts all choice-school utilities in the same direction, implying symmetric substitution patterns between choice schools. Second, we make assumptions on θ_i 's distribution $F(\theta_i)$ to make the model tractable. We approximate $F(\theta_i)$ via a finite mixture of K -many normals. Third, as we detail further below in our model of achievement, θ_i is allowed to affect both achievement levels and the gains from choice-school enrollment.

Application Decisions The application decision depends on preferences for schools outlined in Equation (3), the probability of receiving offers, and application costs. Consistent with LAUSD application rules in place from 2001 to 2013, each family is permitted to apply to a single program.¹⁵ The application vector is $A_i = (A_{i1}, \dots, A_{iJ})$ where A_{ij} is the indicator for whether student i applies to school j . Similarly, the offer vector is $Z_i = (Z_{i1}, \dots, Z_{iJ})$ where the indicator Z_{ij} denotes receipt of an offer. As noted above, offers are realized from a lottery embedded in the centralized assignment system conditional on student i applying and school j being oversubscribed. Students face different probabilities, denoted π_{ij} , of receiving a lottery-based offer to school j based on priority groups defined based on sibling status, race, and neighborhood proximity. Given the uncertainty in admissions, the expected utility from a choice set that includes an offer from j , $Z_{ij} = 1$, is:

$$u(Z_{ij} = 1 \mid X_i, D_i, \theta_i) = \log \left(1 + \exp((\delta_j + \theta_i) + \beta_c X_i - (\psi_D D_{ij} + \psi_{D2} D_{ij}^2 + \psi_{Dx} D_{ij} \times X_i)) \right) + \gamma, \quad (4)$$

where γ is Euler's constant. Therefore, the ex-ante expected utility from applying to j is:

$$e(A_{ij} = 1 \mid X_i, D_i, \theta_i) = \pi_{ij} u(Z_{ij} = 1 \mid X_i, D_i, \theta_i) + (1 - \pi_{ij}) \gamma. \quad (5)$$

Students weigh expected utility from different choices of application vectors against their application costs. Let $a = (a_1, \dots, a_J)$ denote a possible choice school application vector. The utility cost of submitting this set of application choices is given by $c(a, X_i, \eta_i)$ and η_i governs unobserved application cost heterogeneity. We assume $\eta_i \sim N(0, \sigma_\eta^2)$, and the realizations of η_i are known to each student. Application costs are assumed to be:

¹⁵Our baseline model reflects LAUSD's historical one-application regime, but we relax this restriction in the "decentralized" counterfactual scenario (Section 9), where families are permitted to submit multiple applications.

$c(a, X_i, \eta_i) = \mathbf{1}\{\sum_j a_j > 0\} \times (\exp(\mu_c + X_i \mu_{cx}) + \eta_i)$, where the parameters μ_c and μ_{cx} are the mean application cost common to all applications and the heterogeneous application costs with respect to observables, respectively. The unobserved application cost η_i is incurred when a student chooses to apply to the given school. The functional form of application costs allows us to separately identify preference heterogeneity from cost heterogeneity.

Putting this together, students select an application vector $A_i \in \mathcal{J} \cup \{0\}$ such that:

$$A_i = \arg \max_a \sum_j \left(e(a_j = 1 \mid X_i, D_i, \theta_i) \right) - c(a, X_i, \eta_i). \quad (6)$$

Equation (6) does not have a closed form expression, so we follow standard approaches in approximating the choice probabilities using a logit kernel smoother. For a small λ , we approximate the application choice probabilities as:

$$P_A(A_i = a \mid X_i, D_i, \theta_i, \eta_i) \approx \frac{\exp \left(\left\{ \sum_j e(a_j = 1 \mid X_i, D_i, \theta_i) - c(a, X_i, \eta_i) \right\} / \lambda \right)}{\sum_{a'} \exp \left(\left\{ \sum_j e(a'_j = 1 \mid X_i, D_i, \theta_i) - c(a', X_i, \eta_i) \right\} / \lambda \right)}, \quad (7)$$

where a' is the index over the set of possible application vectors.

Enrollment Decisions When students submit applications, they subsequently receive an offer set $\mathcal{O}_i$ that includes the single choice school to which they applied and their default neighborhood school; students who did not apply have only their neighborhood school in $\mathcal{O}_i$. We introduce a post-lottery Type I extreme value preference shocks, ξ_{ij} , which captures the possibility that students may apply to a choice school, receive an offer, yet decline enrollment. Conditional on the realized offer set and the vector of covariates $(Z_i, X_i, D_{ij}, \theta_i)$, the probability that student i enrolls in school j —conditional on receiving an offer—is therefore given by a standard logit over the available options:

$$P_E(E_i = j \mid Z_{ij} = 1, X_i, D_i, \theta_i) = \frac{\exp(v_j(X_i, D_i, \theta_i))}{\sum_{j' \in \mathcal{O}_i} \exp(v_{j'}(X_i, D_i, \theta_i))}, \quad (8)$$

where $v_j(\cdot)$ denotes systematic utility. Equations (7) and (8) demonstrate that observable and latent tastes for choice schools, θ_i , influence application and enrollment decisions. In

the achievement outcome model specified below, it is this model-implied variation that we use to account for selection into both application and eventual enrollment.

Estimating Preference and Cost Parameters Equations (7) and (8) imply a likelihood contribution for student i who submits application A_i , receives offer vector Z_i , and enrolls in school E_i . In our main analysis, we assume that θ_i follows a K -component normal mixture, while η_i is normally distributed:

$$dF(\theta, \eta \mid X_i, D_i) = \left(\sum_{k=1}^K p_k \phi(\theta; \mu_k, \sigma_k^2) \right) \phi(\eta; 0, \sigma_\eta^2) d\theta d\eta,$$

where p_k denotes the share of type k and $\sum_{k=1}^K p_k = 1$. Integrating over these unobservables, the individual likelihood contribution is:

$$\begin{aligned} \mathcal{L}(A_i, Z_i, E_i \mid X_i, D_i) = & \sum_{k=1}^K p_k \int P_A(A_i \mid X_i, D_i, \theta, \eta) P(Z_i \mid A_i, X_i) \\ & \times P_E(E_i \mid Z_i, X_i, D_i, \theta) \phi(\theta; \mu_k, \sigma_k^2) \phi(\eta; 0, \sigma_\eta^2) d\theta d\eta, \end{aligned} \quad (9)$$

where $P(Z_i \mid A_i, X_i)$ is the probability mass function of offers conditional on applications. We maximize the simulated log likelihood function corresponding to Equation (9) using all students in the sample. We recover a vector of parameters, $\Omega = (\{\delta_j\}, \beta_x, \psi_D, \psi_{D2}, \mu_c, \mu_{cx}, \{\mu_k, \sigma_k, p_k\}_{k=1}^K, \sigma_\eta)$. To resolve the arbitrary labeling of the mixture components, we order types in terms of increasing choice school preference, so that $\mu_k > \mu_{k'}$ implies $k > k'$. See Supplemental Appendix C for details on the demand model and estimation.

Identification of Demand Parameters Identification of our choice school demand model relies on three features of the data. First, enrollment decisions depend on the randomized choice-school offers Z_i , which are exogenous to unobserved tastes and application costs. The offer timing implies that Z_i is excluded from the application decision and helps identify the mean utilities δ_j from observed enrollment patterns. Variation in Z_i within lottery strata introduces variation in students' feasible choice sets, allowing the model to trace how take-up varies across schools and thereby recover mean utilities and other demand parameters. Cross-sectional differences in enrollment behavior identify β_c by comparing how choice school demand varies with demographics and baseline achievement, while the

degree to which otherwise similar students vary enrollment with respect to changes in distance identifies the travel-cost parameters (e.g., ψ_D and ψ_{D2}).

Second, application costs in the baseline version of our model, where students submit a single application, are identified by extensive margin variation. Conditional on the enrollment preferences, the application decision depends on the expected utility of enrolling at school j and the known offer probabilities π_{ij} . Naturally-occurring variation in admission probabilities make it so that observationally similar students—with similar preferences—make different application decisions, allowing us to separately identify application costs from preferences (e.g., [Fu, 2014](#)).¹⁶

Third the idiosyncratic parameter θ_i , which acts as a common taste for choice schools, is disciplined by the joint behavior of application and enrollment decisions. Stronger substitution toward choice schools at enrollment and higher propensities to apply when expected option values rise both load onto θ_i . This is a standard finite-mixture/random-coefficients identification argument: multiple observed margins that share a latent factor help recover its location dispersion. In our $K > 1$ specification, the mixture parameters $\{\theta; \mu_k, \sigma_k^2\}$ are recovered from heterogeneous responsiveness on the enrollment and application margins, while distance identifies travel disutility, so θ_i is not correlated with geography.

7.3. Student Achievement Model

This section lays out a model of causal effects of choice schools for students throughout the district. Our framework assumes students choose schools optimally, implying that students at a given school are not a random sample of the population. We address selection by allowing mean potential outcomes to depend on the same unobserved preference, θ_i , that governs application and enrollment decisions. Specifically, we assume:

$$E[Y_{ij} | X_i, \theta_i] = \alpha_j + \gamma'_0 X_i + \gamma'_c X_i \times \mathbf{1}\{j > 0\} + g_j(\theta_i) \quad j = 0, 1, \dots, J, \quad (10)$$

¹⁶Following the transparency diagnostics advocated by [Andrews et al. \(2020\)](#), Appendix Figure [A.8](#) perturbs the fixed-cost estimate and shows how predicted application rates move away from the observed rate; the exercise is analogous in spirit to [Autor et al. \(2019\)](#).

where the indicator $\mathbf{1}\{j > 0\}$ denotes whether school j is a choice program, and $g_j(\cdot)$ captures the relationship between unobserved choice-school preferences and potential achievement outcomes.

Our approach imposes the following restrictions. First, we exclude Z_i from the achievement potential outcomes, a standard approach in the school choice literature. Second, we exclude D_i from the achievement equation, so that residential proximity influences achievement only through its role affecting application and enrollment decisions. This restriction parallels the use of distance instruments in the charter school literature and rules out direct effects of commuting distance on test scores conditional on school attendance. In our robustness exercises, we relax this distance exclusion restriction by allowing distance to enter the achievement equation directly, and use busing-eligibility indicators as the excluded shifters for enrollment (see near the end of Section 8). Third, we assume application costs $c(a, X_i, \eta_i)$ are unrelated to potential outcomes, so that unobserved cost heterogeneity influences who enters the applicant pool but not academic achievement directly. Fourth, we exclude the post-lottery preference shocks ξ_{ij} from potential outcomes.

These restrictions imply that the remaining selection into choice schools operates through the unobserved preference θ_i . In our main approach, we parameterize its relationship with potential outcomes linearly following prior work ([Abdulkadiroğlu et al., 2020](#), [Walters, 2018](#), [Bruhn et al., 2023](#), [Otero et al., 2023](#)). Specifically, we assume:

$$g_j(\theta_i) = \gamma_\theta \theta_i + \gamma_{c\theta} \theta_i \times \mathbf{1}\{j > 0\}. \quad (11)$$

In this model, the parameters γ_θ and $\gamma_{c\theta}$ capture selection on achievement levels and on treatment gains, respectively. If $\gamma_\theta > 0$, then students with stronger latent preferences for choice schools tend to have higher achievement regardless of their eventual enrolled school. In the case that $\gamma_{c\theta} > 0$, students with stronger choice-school preferences experience larger achievement gains from enrollment.

Our assumptions imply that the relevant control function is the posterior mean of the latent choice-school preference: $\theta_i^* \equiv E[\theta_i \mid X_i, D_i, A_i, Z_i, E_i]$. It follows that $E[g_j(\theta_i) \mid X_i, D_i, A_i, Z_i, E_i] = \gamma_\theta \theta_i^* + \gamma_{c\theta} \theta_i^* \times \mathbf{1}\{E_i > 0\}$. We calculate θ_i^* by simulation after estimating the demand model. The estimates of θ_i^* serve as a control function for the non-random sorting of students. Conditioning on this posterior mean accounts for the

unobserved preference heterogeneity that jointly influences application, enrollment, and achievement under the maintained assumptions.

As an alternative, we also consider a type-based specification, which replaces $g_j(\theta_i)$ with a function h_j of the latent mixture type and allows both achievement levels and treatment effects to vary across the types used to approximate the distribution of θ_i . Formally, let $T_{ik} = 1(\tau_i = k)$ with $\sum_k T_{ik} = 1$. Our parameterization assumes: $h_j(\tau_i) = \sum_{k \neq 1} \gamma_k T_{ik} + \sum_{k \neq 1} \gamma_{ck} T_{ik} \times \mathbf{1}\{j > 0\}$. The parameters γ_k and γ_{ck} play the same roles as γ_θ and $\gamma_{c\theta}$ in the linear specification, capturing selection on achievement levels and treatment gains, respectively, but allowing these relationships to vary flexibly across latent types. Because type membership is unobserved, the corresponding control functions are the posterior type probabilities: $p_{ik}^* = E[T_{ik} | X_i, D_i, A_i, Z_i, E_i]$, which we calculate from the estimated demand model.

Having constructed the corresponding control functions, we estimate both achievement specifications, augmented with neighborhood and year fixed effects. Importantly, neither specification is restricted to lottery applicants, and both account for selection into application and enrollment, allowing us to characterize treatment-effect heterogeneity for the full district population.

Identification of the Student Achievement Parameters Identification of the student achievement model relies on variation in lottery offers and relative distance. Lottery offers provide exogenous variation in who enrolls, conditional on applying, while variation in distance changes the composition of the applicant pool. A key concern with the latter source of variation is that cross-sectional distance may correlated with student or neighborhood characteristics that predict achievement. Given this, our approach instead relies on policy-induced changes in proximity generated by LAUSD's expansion of choice programs, which distinguishes our approach from [Walters \(2018\)](#)'s approach to studying charter schools. We implement this strategy by including neighborhood and year fixed effects in our achievement model.

While Section 5.1 establishes relevance by demonstrating that choice school entry increases local application rates, we now assess whether the same shocks are accompanied by changes in observable student composition that could confound their interpretation as demand shifters. Appendix Figure A.7 examines the observable composition of the appli-

cants drawn in by expansion. We construct a tract-level composition index by predicting student-level achievement from baseline covariates and averaging the fitted values among applicants in a tract. Re-estimating the stacked event study with this index and its components as outcomes shows flat pre-trends and no meaningful post-entry shifts in the index. In sum, expansions tend to draw in more applicants while preserving the compositional mix of applicants.

A complementary exercise asks whether the changes in access are directly associated with changes characteristics. Figure 4 shows that cross-sectional distance is moderately correlated with several key student observables (e.g., race and poverty measures). However, once both year and neighborhood fixed effects are included—so that identification comes from within-neighborhood changes in proximity—the estimated correlations largely shrink and are uniformly small. This pattern confirms that variation in access induced by new school openings is not predicted by observable student characteristics.

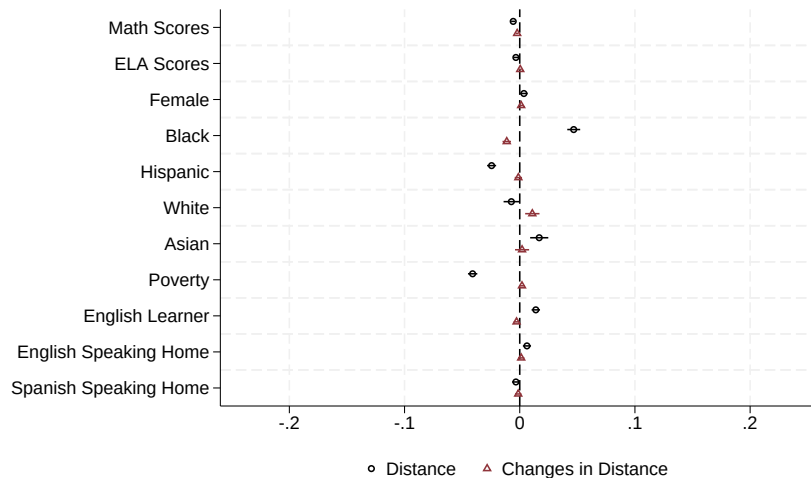


FIGURE 4.—Correlations Between Choice School Distance and Student Characteristics. *Note:* This figure reports coefficients from student-level regressions of baseline characteristics on relative distance to the nearest choice school, defined as a block's distance to the nearest choice program minus its distance to the nearest non-choice neighborhood school. The sample includes all fifth-grade students from 2004 through 2017. Circle markers correspond to specifications with academic-year fixed effects; triangle markers include neighborhood and year fixed effects, isolating within-neighborhood changes in distance. Bars report 95% confidence intervals using robust standard errors clustered by neighborhood.

8. MODEL ESTIMATES AND CAUSAL EFFECTS

This section begins by reporting the structural model estimates. We then summarize causal effects for the entire population of LAUSD students, uncovering some of the allocative inefficiencies of opt-in systems. We conclude by analyzing the alignment of school-specific demand and school-specific causal effects and report results from a set of validation and robustness checks.

Model Fit We estimate three model versions that vary in the number of components K in the mixture of normals used to approximate the distribution of unobserved preference heterogeneity. Appendix Table A.6 reports model fit statistics and shows that the log-likelihood is maximized with a three-type model ($K = 3$), although the incremental gain relative to the two-type specification is modest. Further improvement in fit beyond the three-type model is also modest, so the three-type model is our preferred specification.

We next conduct an out-of-sample validation to assess the overall fit of our preferred model that uses a three-type mixture distribution. Our approach estimates the model in a 25% random training sample and uses the remaining 75% to assess model fit.¹⁷ Appendix Figure A.9 compares school-level aggregates and group-by-school-level aggregates. The results show that the model is able to match school-specific application demand closely. For applications, the model forecasts school-specific demand accurately on average with a forecast coefficient equal to 0.99 (std. err. = 0.043). The figure also shows similarly strong performance for enrollment, with a forecast coefficient equal to 0.97 (std. err. = 0.034). For both stages, we find that the model predictions of application and enrollment behavior are forecast unbiased.

Demand Model Estimates Appendix Table A.7 summarizes our demand model estimates. Our discussion focuses on three features of the estimates that are most important for our analyses that follow: (i) the magnitude of travel and application costs; (ii) the relative attractiveness of choice and neighborhood schools; and (iii) heterogeneity in students' demand for choice schools.

¹⁷We run 200 simulations in which we use parameter estimates from the training sample to simulate applications, offers, and enrollment in the holdout sample.

First, the estimates indicate that both travel distance and application costs shape participation. Consistent with prior evidence on the importance of travel distance for school choice, the estimated effect of relative distance on demand is negative and statistically significant. Families also face a fixed cost of submitting an application. For the typical student, the estimated fixed application cost is equivalent to roughly 3.8 miles of additional travel when scaled by the linear distance-cost coefficient. Thus, the estimates imply that the fixed cost of applying captures a substantial additional barrier beyond travel costs.

Second, the estimates imply that the average student views a typical choice school, net of application and distance costs, as less attractive than their neighborhood-school. The mean choice-school utility intercept is -0.897 , equivalent to approximately 2.75 miles of additional travel. At the same time, demand varies substantially across choice programs: the standard deviation of the school-specific utility is equivalent to approximately 3.2 miles of travel. This implies that some schools are so attractive that, absent travel or application costs, virtually all students would apply.

Third, there is substantial heterogeneity across students in demand for choice schools. Most importantly for the analysis that follows, the estimated mixture distribution reveals large differences in latent choice-school preferences. Roughly 45% of students are assigned to a low-demand type (with mean preference $\mu_1 = -0.876$), 49% to a moderate-demand type ($\mu_2 = 0.430$), and 7% to a high-demand type ($\mu_3 = 2.75$). Demand also varies systematically with observable student characteristics. Students with higher baseline achievement and those already enrolled in choice programs have stronger estimated preferences for choice schools, while Black and Hispanic students and students with limited English proficiency have weaker estimated preferences. These differences in latent and observable demand motivate examining whether students with stronger preferences for choice schools also experience systematically different achievement gains from enrollment.

Causal Effects on Student Achievement Table II reports the selection-corrected parameter estimates based on the student achievement model specified by Equation (10).¹⁸ We report estimates for math and ELA achievement separately.

¹⁸Appendix Table A.9 reports bootstrapped estimates that take into account estimation error in the estimated control functions. Results from the bootstrap exercise are quantitatively similar to our main results.

The results in the top row show that attending an average choice school boosts test scores by 0.25σ in Math and 0.23σ in ELA for the average LAUSD student. The corresponding across-school standard deviations are 0.16σ and 0.11σ , implying substantial vertical differentiation: some programs generate large gains, while others perform similarly to or worse than neighborhood schools. Treatment effects also vary systematically with student characteristics. Black students experience relatively larger gains, Asian students relatively smaller gains, and gains decline with baseline achievement. These patterns closely mirror the reduced-form evidence in Appendix Figure A.6, which showed that characteristics associated with lower application rates tend to predict larger benefits from choice-school attendance.

TABLE II
STUDENT ACHIEVEMENT MODEL ESTIMATES

	Linear Control Function			
	Math		ELA	
	γ_0 (1)	γ_c (2)	γ_0 (3)	γ_c (4)
<i>Panel A. Average choice effects</i>				
Main effects		0.252 [0.158]		0.226 [0.111]
<i>Panel B. Demographic and socioeconomic controls</i>				
Female	-0.022 (0.002)	-0.010 (0.007)	0.111 (0.002)	0.001 (0.006)
Black	-0.197 (0.011)	0.028 (0.022)	-0.161 (0.010)	0.027 (0.020)
White	-0.004 (0.012)	-0.033 (0.021)	0.001 (0.009)	-0.030 (0.021)
Hispanic	-0.150 (0.010)	-0.009 (0.020)	-0.119 (0.008)	0.010 (0.019)
Asian	0.210 (0.014)	-0.059 (0.025)	0.075 (0.009)	-0.041 (0.022)
Poverty	0.010 (0.003)	-0.012 (0.010)	0.003 (0.002)	-0.007 (0.007)
LEP	-0.087 (0.004)	0.041 (0.013)	-0.171 (0.003)	0.012 (0.014)
Median income		-0.006 (0.002)		-0.006 (0.002)
English home	-0.070 (0.004)	0.009 (0.012)	-0.059 (0.004)	0.044 (0.010)
<i>Panel C. Baseline achievement and peer quality</i>				
Baseline Math	0.526 (0.004)	-0.037 (0.009)	0.182 (0.002)	-0.023 (0.006)
Baseline ELA	0.210 (0.003)	0.014 (0.007)	0.591 (0.003)	-0.016 (0.008)
Baseline peer quality	-0.147 (0.012)	0.036 (0.019)	-0.094 (0.009)	0.025 (0.014)
<i>Panel D. Linear control-function parameters</i>				
Choice preference θ_i	0.023 (0.003)	-0.076 (0.009)	0.029 (0.003)	-0.071 (0.007)

Note: Estimates of the observational analog of Equation 10. Odd columns report baseline coefficients (γ_0); even columns report choice-enrollee interactions (γ_c), so the total choice effect is $\gamma_0 + \gamma_c$. “Main effects” are enrollment-weighted average choice effects; noise-adjusted standard deviations are in brackets. All specifications include census-block, year, and subdistrict fixed effects. Robust standard errors clustered by enrolled school are in parentheses.

In terms of idiosyncratic tastes, Table II shows that students with stronger estimated preferences for choice schools are positively selected: a one-unit increase in θ_i is associated with $0.02\text{--}0.03\sigma$ higher achievement regardless of the school attended. At the same time, stronger preferences predict smaller gains from choice-school enrollment. A one-unit increase in θ_i reduces choice-school causal effects by 0.08σ in Math and 0.07σ in ELA.¹⁹ These patterns are consistent with negative selection on achievement gains. The negative selection on achievement gains we find adds to the growing body of evidence of this kind of sorting in education and other settings (Cornelissen et al., 2018, Walters, 2018, Chyn, 2018).

To aid interpretation of these selection patterns, Appendix Figure A.10 summarizes choice-school preferences for who only appear in the lottery sample. We begin by taking the posterior estimates of θ_i for the full population and constructing the associated percentiles. We repeat this procedure for an overall preference index, which is defined as $I_i \equiv \beta_c X_i + \theta_i$. We attach the relevant percentile ranks to the lottery applicants to show where they lie in each distribution. Across both measures, lottery applicants are sharply concentrated in the far right tail: very few fall below the 80th percentile, and a substantial mass lies in the top decile. Combined with the negative relationship between θ_i and treatment gains in Table II, this pattern implies that lottery applicants are disproportionately drawn from students who stand to benefit less from attending choice schools.

Finally, as a complementary way to visualize the treatment effect heterogeneity, Figure 5 reports average choice-school causal effects across the three latent preference types. These results come from the alternative type-specific parameterization of the achievement model. The results show that Type 1 and 2 students, who have weaker tastes for choice schools, have positive and relatively large-in-magnitude treatment effects from choice school enrollment. In contrast, Type 3 students—those with the strongest choice school tastes—have negative effects. Together with Appendix Figure A.10, which shows that lottery applicants are drawn disproportionately from the high-preference tail of the distribution, this figure makes the reverse-Roy pattern transparent. In summary, our analysis of causal effects re-

¹⁹The alternative type-specific specification reported in Appendix Table A.8 yields the same pattern: Type 2 students' causal effects are $0.12\text{--}0.17\sigma$ lower than those of Type 1 students, while Type 3 students' effects are $0.31\text{--}0.38\sigma$ lower.

veals two takeaways: (i) choice schools are vertically differentiated, likely reflecting a competitive response of school districts in response to growing competition, and (ii) there is stark heterogeneity in treatment effects—a finding that alludes to allocative inefficiency in terms of district-level achievement maximization.

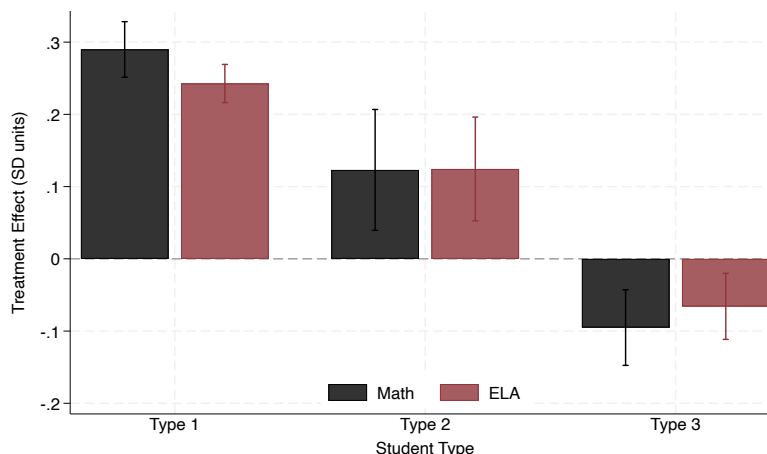


FIGURE 5.—Choice School Causal Effects by Preference Type. *Note:* This figure reports average causal achievement effects for the three latent preference types implied by the demand model, estimated using an alternative type-specific parameterization of the achievement model. Appendix Table A.8 reports the parameter estimates used to construct these type-specific estimates. Bars report 95% confidence intervals using robust standard errors clustered by school and the delta method where appropriate. Math and ELA estimates use 334,166 students.

Demand for School Effectiveness The negative relationship between individual preferences and treatment effects raises a related question: are more effective choice schools more popular? Appendix Figure A.11 plots school-specific causal effects against estimated demand and shows little evidence of a positive relationship; if anything, more effective schools are less popular. This weak alignment between demand and effectiveness is consistent with prior evidence (Abdulkadiroğlu et al., 2020) and may reflect preferences for other school attributes or limited information about effectiveness (Rothstein, 2006, Ainsworth et al., 2023, Campos, 2024).

Robustness and Additional Validation We conduct three exercises that address the assumptions underlying the achievement model. First, we allow both potential outcomes

and treatment effects to vary with neighborhood-school value added. Appendix Table A.10 shows that students with stronger outside options experience smaller choice-school gains, but accounting for this heterogeneity leaves our estimates of average effectiveness and negative selection on gains essentially unchanged. Second, as previewed above, we relax the exclusion restriction on distance by allowing distance to enter the achievement model directly and instead use the district's three-mile busing-eligibility threshold, together with lottery offers, as excluded enrollment shifters. Appendix Table A.11 again yields similar estimates, indicating that direct effects of distance do not explain our selection results.²⁰ Third, we validate the model against lottery-based causal effects following Angrist et al. (2017). Appendix Figure A.12 compares each lottery's effect on model-predicted achievement with its reduced-form effect on observed achievement. The resulting forecast coefficient is 0.91 and is statistically indistinguishable from one, indicating that the model predicts causal effects commensurate with those identified experimentally.

9. COUNTERFACTUAL POLICY ANALYSIS

Our findings to this point suggest that participation in LAUSD's choice system remains limited and uneven, with heterogeneity in who applies and how much schools contribute to student achievement. Alternative policies and system designs could potentially reshape achievement by changing how families access and engage with choice programs. We explore these possibilities using the estimated structural model, adjusting parameters governing application, enrollment, and achievement.

For our policy analysis, LAUSD's current opt-in structure serves as a baseline simulation, providing a benchmark for comparisons with scenarios that incorporate targeted participation-focused interventions or system-level reforms. The participation policies address key barriers to entry: (i) information interventions that improve families' understanding of available options, modeled as exogenous increases in latent demand for choice schools through θ_i ;²¹ and (ii) reductions in travel costs, such as universal busing, implemented as reductions in the distance-cost parameters. The system-level reforms are (iii)

²⁰Prior studies exploiting transportation and assignment policies likewise find small test-score effects of travel time (e.g. Angrist et al., 2022, Cordes et al., 2022, Edwards, 2023).

²¹Changes in participation architecture could also encourage information acquisition, a channel implicitly held fixed in most counterfactuals. This channel is unlikely to be quantitatively important: pairing mandatory partic-

mandatory participation, as in New York City and Boston, modeled by setting application costs to zero; and (iv) decentralized choice, in which families may apply to multiple schools and receive multiple offers in the absence of a centralized clearinghouse. Collectively, our simulations capture the dominant system designs adopted by nearly all large districts and incorporate participation-enhancing policies, such as busing and information interventions, that many districts have implemented.

On the supply side, we fix the set of choice schools, their locations, and their school-level utilities (δ_j) and achievement effects (α_j). We also hold fixed each program's race-specific seat capacities and run admissions within program-year-grade-priority points-race strata, mirroring LAUSD's implementation.²² Maintaining these targets ensures that changes arise from family decisions rather than the relaxation of integration mandates. For each policy, we calculate equilibrium admission probabilities under the resulting demand structure. These assumptions isolate how family decision-making, rather than school expansion or reorganization, affects participation and achievement. Appendix C provides further details.

We begin by examining information interventions in the context of the existing opt-in system. Table II suggests that inducing students with lower θ_i to apply can increase system-wide learning. We simulate two common information-provision strategies (Hastings and Weinstein, 2008, Ainsworth et al., 2023). Both replace θ_i with $\theta'_i = \theta_i + \Delta_\theta$ in application and enrollment decisions while retaining the original θ_i when calculating achievement. In the initial general information experiment, we set Δ_θ to a type-specific one standard deviation (reported in Column 2 of Table A.7) for a randomly selected subset of students, capturing imperfect compliance and heterogeneity in take-up. In the next, school-targeted experiment, we focus only on students enrolled in low-achieving schools, defined as those with average achievement below the district average, and apply the same Δ_θ increase to θ'_i for a random subset within this group to mimic targeted outreach. Together, these scenarios

ipation with increased information produces results essentially indistinguishable from mandatory participation alone.

²²As noted in Section 4, choice programs have used racial enrollment quotas that have historically reserved a large share of seats for students from PHBAO (Predominantly Hispanic, Black, Asian, and Other non-Anglo) neighborhoods; these requirements persisted after court supervision ended and were upheld as lawful given their pre-1996 remedial origin.

ios illustrate how information provision could shift participation and sorting in LAUSD's choice system by narrowing informational gaps across families.

Table III shows that both information interventions produce extremely modest achievement gains, raising district-level achievement by 0.003σ in Math and 0.002σ in ELA.²³ Columns 1 and 2 provide additional insight, demonstrating that application rates increase from 16-20% and the number of occupied seats by 11-12 percentage points from a base of 55.3%. By demographic group, Asian and White students have the largest enrollment responses in percentage-point terms to information campaigns, as has been shown in prior reduced form work (Corcoran et al., 2018). The skewed take-up rates produce minimal impacts on district-level achievement, given that the affected students induced to participate have relatively weaker achievement gains (see Table II and Appendix Figure A.14). Information campaigns can therefore reallocate students across schools, but do not overcome the substantial participation barriers inherent in opt-in systems.

We next evaluate expanded busing as a way to reduce travel barriers. Busing policies have a long history of broadening access to schools.²⁴ Prior research on busing policies largely finds that travel distance has little impact on achievement (e.g. Angrist et al., 2022, Cordes et al., 2022, Edwards, 2023), which helps alleviate concerns that commuting burdens directly drive student outcomes in our setting. At the same time, travel costs are central in our model of school choice: we estimate sizable disutility from distance, implying that improved busing availability or salience could meaningfully affect enrollment, particularly among disadvantaged students.

Table III shows that eliminating travel costs raises district-level achievement by 0.009σ in Math and 0.007σ in ELA, with larger effects for white students.²⁵ These gains are associated with 33% of families participating and 86% of seats being occupied, substantial increases relative to the baseline opt-in system. Consistent with reductions in travel costs drawing in relatively disadvantaged students, the effects on the set of enrolled students (i.e., the treatment on the treated) increase by 4% of a standard deviation for both Math and

²³Appendix Figure A.14 provides a graphical comparison of their district-level achievement effects.

²⁴In Los Angeles, busing was once part of court-ordered integration efforts, and today the district operates a voluntary busing program that facilitates access to magnet programs.

²⁵See Appendix Figure A.14 for race-specific results for each counterfactual considered in this section.

ELA. These estimates provide an encouraging benchmark, although the simulation does not account for the potentially substantial cost of eliminating travel costs (Angrist et al., 2022).

Because access-targeting leaves substantial non-participation, we now assess two system-level alternatives. We first consider a decentralized alternative to LAUSD’s coordinated opt-in system. Families submit multiple applications and schools run independent lotteries, potentially yielding multiple offers. We introduce a marginal per-application cost to discipline portfolio size in the model. The ability to submit multiple applications has the potential to thicken the market and reduce slack, but also creates congestion that may discourage marginal entrants. Second, under a mandatory participation design, every student submits a rank-ordered list, and a deferred-acceptance clearinghouse coordinates offers using uncapped list lengths. We set application costs to zero, although families continue to weigh preferences and travel costs when ranking schools. By removing application frictions and multi-offer congestion, mandatory participation can broaden access, fill capacity, and reallocate students toward higher-gain seats. Prior work finds that transitions to coordinated assignment can improve welfare (Abdulkadiroğlu et al., 2017). Thus, decentralization adds congestion, while mandatory participation broadens access and coordinates assignment.

Despite having larger impacts on choice school seat utilization, Table III shows that the decentralized system yields only small gains in district achievement that comparable in size to information interventions. As families expand their application portfolios, equilibrium admission probabilities fall, which induces some marginal families to stop applying and overall participation drops by about 5 percentage points. Among those who remain, most still secure at least one offer, so capacity is used more fully relative to the opt-in baseline. The composition of applicants shifts toward families with lower application costs, producing a more advantaged pool and smaller treatment-on-the-treated (TOT) effects for ELA scores, even as district averages tick up slightly. In short, decentralization discourages entry overall and reallocates seats toward inframarginal applicants, yielding small district gains.

Our next set of system-level results shows that a transition to a mandatory system with a deferred acceptance mechanism for assignment raises achievement levels by roughly 0.02σ in both Math and ELA. The TOT effects for choice school enrollees also increase substantially by about 0.10σ . These impacts are driven by a 69.1 percentage point increase in the

share of all families applying to a non-neighborhood school as their most-preferred option (recall that “apply” is defined as ranking a non-neighborhood school first; families who comply with the mandate by listing only their neighborhood school are counted as non-applicants). This evidence suggests that application costs are the empirically dominant barrier to participation under opt-in systems. The mandatory system also eliminates essentially all excess choice-school capacity, with 98% of seats occupied compared to 55% at baseline. It reallocates students across schools, reaches students with larger causal effects, and raises district-level achievement.

Having demonstrated that a mandatory system with deferred acceptance produces the largest system-level gains, we move to examine whether combining mandatory participation with other policies produces additional gains. Adding the generous busing policy raises the share of families applying to a choice school to 99%, a sizable increase relative to mandatory participation alone. Although the bump in demand is sizable, this has only a modest change in the number of seats filled, given that capacity constraints are essentially binding in the case of mandatory participation. Consequently, district-wide and TOT achievement gains are marginally changed relative to the mandatory-only case. Simulations that pair the information campaign with mandatory participation (with or without busing) deliver similarly small incremental gains, reflecting the same capacity constraint.

Finally, the simulations above show that several implementable interventions raise achievement, which naturally prompts the question: can a district do even better? We therefore compute an “achievement-optimal” allocation in which a fully informed planner assigns students solely to maximize district-wide achievement. Following [Abdulkadiroğlu et al. \(2025\)](#), we call this the treatment effect maximizing allocation: it is the solution of a linear program that maximizes achievement subject to district imposed capacity constraints. This “achievement-optimal” allocation delivers average gains of 0.038σ in Math and 0.034σ in ELA at the district level, fills all remaining choice capacity (moving from 55% filled at baseline to nearly 100%), and raises treatment-on-the-treated effects to 0.37σ (Math) and 0.32σ (ELA). Mandatory participation therefore generates roughly half the gains of the infeasible achievement-optimal policy. Demand for school effectiveness continues to be a limiting factor ([Abdulkadiroğlu et al., 2020](#), [Rothstein, 2006](#), [Ainsworth et al., 2023](#)).

While the counterfactuals show that changing participation architecture can increase district-wide achievement, there are at least three concerns. First, because the policies re-allocate students across schools, peer effects could affect the conclusions. We address this by plotting reduced-form school-specific achievement effects against their lottery-induced effects on peer quality (Appendix Figure A.13). If peer quality is an important driver of student outcomes, we would expect a strong correlation between these two variables. Instead, we find no substantive relationship between changes in peer quality and own-achievement effects. Second, the policy changes could generate school-level demographic compositions that lie far outside the observed support. Appendix Figure A.15 shows that each counterfactual produces only modest changes in the spatial distribution of enrolled student demographics. Third, although Appendix Tables A.10 and A.11 demonstrate that choice school treatment effects and treatment effect heterogeneity are robust to accounting for variation in neighborhood school quality and alternative sources of identifying variation, it is less clear how their inclusion affects the system level effects we document. Appendix Tables A.12 and A.13 report counterfactual estimates using the alternative demand and outcome model estimates. The results are all qualitatively and quantitatively similar to our preferred estimates in Table III. Thus, peer effects, demographic reallocation, and richer demand and treatment-effect heterogeneity are unlikely to overturn our main findings.

Why does mandatory participation produce the largest benefits relative to opt-in or other policies? The answer is that choice schools offer meaningful achievement gains, but the baseline opt-in system exhibits two allocative inefficiencies in how those gains are distributed. In Los Angeles and nationally, opt-in systems disproportionately attract academically and socioeconomically advantaged families, students who potentially stand to gain the least from access to more effective schools. In our analysis of LAUSD, our estimates further show that students with stronger demand for choice schools tend to experience smaller achievement gains from attending them. Application costs, limited information, and other participation barriers therefore screen out many students with larger potential gains. Second, these same barriers leave substantial capacity at effective schools unused. Mandatory participation addresses both inefficiencies: eliminating application barriers broadens participation toward students with larger gains, while coordinated assignment fills seats that would otherwise remain empty.

TABLE III
ANALYSIS OF PARTICIPATION REFORMS AND ALTERNATIVE CHOICE SYSTEMS

Panel (a): Baseline Simulation

	Participation (%)		All Student Achievement (σ)		Choice Enrollee TOT (σ)		Demographic Group Choice Enrollment (%)				
	Apply	Choice Seats Filled	Math	ELA	Math	ELA	Asian	Black	Hispanic	White	Low SES
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Opt-in (Voluntary)	15.72	55.33	.	.	0.111	0.078	17.47	7.68	5.01	14.36	5.78

Panel (b): Alternative Policies/System Impacts (Relative to Baseline)

	Participation Gains (%)		All Student Achievement Gains (σ)		Choice Enrollee TOT Gains (σ)		Demographic Group Choice Enrollment Gains (%)				
	Apply	Choice Seats Filled	Math	ELA	Math	ELA	Asian	Black	Hispanic	White	Low SES
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Broad Information	+ 3.84	+ 11.43	+ 0.003	+ 0.002	+ 0.015	+ 0.014	+ 2.21	+ 2.19	+ 1.15	+ 2.05	+ 1.27
Targeted Information	+ 3.90	+ 11.85	+ 0.003	+ 0.002	+ 0.012	+ 0.012	+ 1.54	+ 2.31	+ 1.29	+ 1.67	+ 1.40
Busing Program	+ 17.45	+ 30.33	+ 0.009	+ 0.007	+ 0.042	+ 0.040	+ 1.56	+ 3.35	+ 3.26	+ 9.35	+ 3.83
Decentralized	-5.28	+ 18.53	+ 0.003	+ 0.002	+ 0.001	-0.001	+ 7.09	+ 4.01	+ 1.45	+ 4.72	+ 1.87
Mandatory App.	+ 69.12	+ 42.49	+ 0.018	+ 0.016	+ 0.105	+ 0.103	-14.53	+ 3.06	+ 4.94	+ 15.74	+ 5.33
Mandatory App. + Busing	+ 83.07	+ 44.39	+ 0.021	+ 0.019	+ 0.123	+ 0.121	-15.00	+ 2.29	+ 5.21	+ 18.41	+ 5.61
Mandatory App. + Broad Info	+ 70.00	+ 42.76	+ 0.018	+ 0.016	+ 0.106	+ 0.104	-14.55	+ 2.99	+ 4.97	+ 16.05	+ 5.35
Mandatory App. + Broad Info + Busing	+ 83.17	+ 44.41	+ 0.021	+ 0.019	+ 0.124	+ 0.121	-15.01	+ 2.27	+ 5.22	+ 18.43	+ 5.61
Maximized Achievement	–	+ 44.67	+ 0.038	+ 0.034	+ 0.258	+ 0.245	-16.67	+ 2.99	+ 5.88	+ 18.83	+ 5.41

Note: Simulated results from the application, enrollment, and achievement model. Panel A reports the opt-in baseline; Panel B reports counterfactual differences. Columns 1–2 report first-choice applications and seats filled; Columns 3–4 district-wide achievement; Columns 5–6 treatment-on-the-treated effects; and Columns 7–11 choice enrollment by group. Information policies raise θ_i for 50% of the relevant population; No Travel Costs sets travel costs to zero; Decentralized Choice permits multiple applications and school-run lotteries; and Mandatory Participation eliminates application costs and uses deferred acceptance. Each result uses 100 simulations and, when school-run lotteries remain, equilibrium admission probabilities.

10. CONCLUSION

Our analysis of LAUSD’s opt-in system demonstrates the promise and the limits of expanding school choice in public school districts. We provide evidence that public choice schools are vertically differentiated and, on average, deliver meaningful achievement impacts. Yet, market design details—optional participation in particular—ultimately mutes potential benefits. Because application costs deter many students, especially those who would gain the most from attendance, effective schools often operate below capacity. At the same time, the students who do navigate the application process are disproportionately advantaged. These twin features of the opt-in system contribute to both inequality in educational outcomes and allocative inefficiencies.

The central finding from our counterfactual analysis is that centralized, mandatory assignment can deliver substantial gains. Such reforms reallocate students to more effective schools, fill unused seats, and erode negative selection on gains. Centralization generates achievement improvements roughly half as large as an achievement-maximizing allocation, underscoring the limiting influence of misaligned demand for school effectiveness. However, unlike lighter-touch interventions that provide information or reduce travel costs via generous busing schemes, centralization addresses the dominant frictions of opt-in systems.

Taken together, our results demonstrate that centralization is not a matter of administrative convenience, but a core design feature that mediates school choice effectiveness. More broadly, our findings highlight how institutional features shape the distribution of educational opportunity: expanding public school choice will benefit the students who stand to gain the most only if assignment mechanisms bring those students into the system. Our findings also speak to the potential implications of recent state-level voucher programs, which have expanded opt-in models and raise similar questions about participation and allocation, particularly when the quality of available options is heterogeneous. Overall, our findings demonstrate that system design can improve outcomes even while holding fixed the supply of effective schools.

One question that we cannot answer is why opt-in systems remain so common in large U.S. school districts. A political economy view is that reforms to assignment rules face organized resistance from powerful incumbents and mobilized parent groups, making overhauls costly to attempt and easy to derail—a point developed in the literature on political

constraints in K-12 governance (e.g., [Chubb and Moe, 1988](#)).²⁶ A second, more market-based explanation is that opt-in designs help districts retain families most likely to exit to charters or private schools—typically higher-achieving students seeking high-achieving peers—by letting them self-select into choice without mandating system-wide participation. This incentive is consistent with ongoing debates about cream-skimming and the distributional consequences of decentralized or unified enrollment systems ([Kho et al., 2022](#)). These are, of course, hypotheses. Future work could aim to better understand the relative importance of political versus economic constraints. If opt-in is the binding political equilibrium, the practical challenge is to identify other design margins that can be adjusted to deliver broader gains despite that constraint.

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